

162. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162.1 Overview

This clause specifies the 100GBASE-CR1 PMD, the 200GBASE-CR2 PMD, the 400GBASE-CR4 PMD, and the baseband medium. The specifications for the three PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are four associated annexes. Annex 162A provides information on parameters with test points that may not be testable in an implemented system. Annex 162B specifies test fixtures. Annex 162C specifies MDIs. Annex 162D describes host and cable assembly types.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 162–1, to the appropriate sublayers (as specified in Table 162–1, Table 162–2, and Table 162–3), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 162–1—Physical Layer clauses associated with the 100GBASE-CR1 PMD

Associated clause	100GBASE-CR1
81—RS	Required
81—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC	Required
152—Inverse RS-FEC	Optional ^b
161—RS-FEC-Int	Required
83—PMA for 100GBASE-R	Optional ^c
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135F—100GAUI-2 C2C	Optional
120F—100GAUI-1 C2C	Optional
73—Auto-Negotiation	Required

^a The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^b Inverse RS-FEC is required to convert between RS-FEC and RS-FEC-Int (see 152.1.2).

^c PMA for 100GBASE-R is required when either CAUI-10 or CAUI-4 is used.

Table 162–2—Physical Layer clauses associated with the 200GBASE-CR2 PMD

Associated clause	200GBASE-CR2
117—RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—PCS for 200GBASE-R	Required
120—PMA for 200GBASE-R	Required
120B—200GAUI-8 C2C	Optional
120D—200GAUI-4 C2C	Optional
120F—200GAUI-2 C2C	Optional
73—Auto-Negotiation	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

Table 162–3—Physical Layer clauses associated with the 400GBASE-CR4 PMD

Associated clause	400GBASE-CR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—PCS for 400GBASE-R	Required
120—PMA for 400GBASE-R	Required
120B—400GAUI-16 C2C	Optional
120D—400GAUI-8 C2C	Optional
120F—400GAUI-4 C2C	Optional
73—Auto-Negotiation	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

For the 100GBASE-CR1 PHY, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-10} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 10^{-10} after processing by the FEC (Clause 91 or Clause 161) and the PCS (Clause 82) if there are negligible errors due to other electrical interfaces (100GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 10^{-10} .

For the 200GBASE-CR2 and 400GBASE-CR4 PHYs, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the

adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 9.2×10^{-13} after processing by the PCS (Clause 119) if there are negligible errors due to other electrical interfaces (200GAUI-n or 400GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 9.2×10^{-13} .

A compliant input signal is a transmitter output of a compliant PHY that has passed through a compliant cable assembly.

Figure 162–1 shows the relationship of the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMD sublayers and MDIs to the ISO/IEC Open System Interconnection (OSI) reference model.

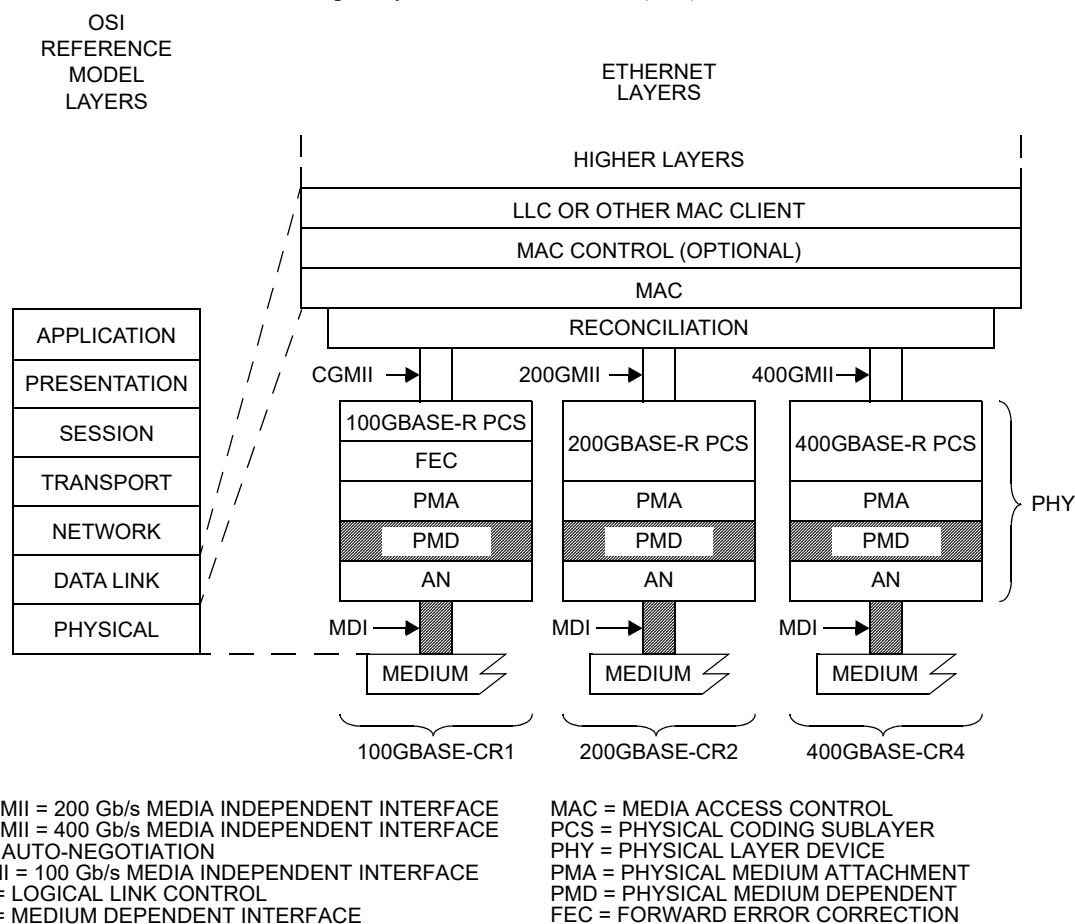


Figure 162–1—100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

162.2 Conventions

Clause 162 describes three PMDs, 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, which have one, two, and four lanes, respectively. For efficient description, the parameter n is used to describe the

number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-CR1, $n = 2$ for 200GBASE-CR2, and $n = 4$ for 400GBASE-CR4.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-CR1 PMD, 200GBASE-CR2 PMD, or 400GBASE-CR4 PMD.

Within this clause, the term FEC refers to either Clause 91 RS-FEC or the Clause 161 RS-FEC-Int for 100GBASE-CR1 or the RS-FEC within the Clause 119 PCS for 200GBASE-CR2 and 400GBASE-CR4.

162.3 PMD service interfaces

This subclause specifies the services provided by the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The 100GBASE-CR1 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with the exception that there is a single symbol stream ($n = 1$).

The 200GBASE-CR2 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with two parallel symbol streams ($n = 2$).

The 400GBASE-CR4 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with four parallel symbol streams ($n = 4$).

The service interface primitives are summarized as follows:

PMD:IS_UNITDATA_ i .request
PMD:IS_UNITDATA_ i .indication
PMD:IS_SIGNAL.indication

The 100GBASE-CR1 PMD has a single symbol stream, hence $i = 0$.

The 200GBASE-CR2 PMD has two parallel symbol streams, hence $i = 0$ to 1.

The 400GBASE-CR4 PMD has four parallel symbol streams, hence $i = 0$ to 3.

In the transmit direction, the PMA continuously sends n streams of PAM4 symbols to the PMD, one per lane, using the PMD:IS_UNITDATA_ i .request primitive, at a nominal signaling rate of 53.125 GBd. The PMD converts these streams of symbols into appropriate signals on the MDI.

In the receive direction, the PMD continuously sends n streams of PAM4 symbols to the PMA, corresponding to the signals received from the MDI, one per lane, using the PMD:IS_UNITDATA_ i .indication primitive, at a nominal signaling rate of 53.125 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable Global_PMD_signal_detect as defined in 162.8.4. When Global_PMD_signal_detect is one, SIGNAL_OK shall be assigned the value OK. When Global_PMD_signal_detect is zero, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_ i .indication parameter is undefined.

162.4 PCS requirements for Auto-Negotiation (AN) service interface

The PCSs associated with the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs are required to support the AN service interface primitive AN_LINK.indication defined in 73.9. (See 82.6 and 119.6.)

100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PHYs may be extended using CAUI-n/100GAUI-n C2C, 200GAUI-n C2C, and 400GAUI-n C2C, respectively, as physical instantiations of the inter-sublayer service interface between devices. If such extension is used, then the AN_LINK(link_status).indication is relayed from the device with the PCS sublayer to the device with the AN sublayer by means at the discretion of the implementer. As examples, the implementer may employ use of pervasive management or employ a dedicated electrical signal to relay the state of link_status as indicated by the PCS sublayer on one device to the AN sublayer on the other device.

162.5 Delay constraints

The sum of the transmit and the receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 162–4. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 162–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-CR1	4 096	8	40.96
200GBASE-CR2	8 192	16	40.96
400GBASE-CR4	16 384	32	40.96

^a One bit time is equal to 10 ps for 100GBASE-CR1, 5 ps for 200GBASE-CR2, and 2.5 ps for 400GBASE-CR4. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-CR1, 2.56 ns for 200GBASE-CR2, and 1.28 ns for 400GBASE-CR4. (See 31B.2 for the definition of pause_quanta.)

Descriptions of overall system delay constraints can be found in 80.4 for 100GBASE-CR1 and in 116.4 for 200GBASE-CR2 and 400GBASE-CR4.

162.6 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

162.6.1 Skew Constraints for 100GBASE-CR1

Skew and Skew Variation are defined in 80.5 and specified at the points SP0 to SP7 shown in Figure 80–9.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by 135.5.3.5. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 80.5. The measurements of Skew and Skew Variation are defined in 89.7.2 with the exception that the measurement clock and data recovery unit high-frequency corner bandwidth is 4 MHz.

162.6.2 Skew Constraints for 200GBASE-CR2 and 400GBASE-CR4

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

162.7 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the PMD. If MDIO is implemented, it shall map MDIO variables and registers to PMD variables as shown in Table 162–5 through Table 162–7.

Table 162–5—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable	1.9.0	Global_PMD_transmit_disable
PMD transmit disable 3 to PMD transmit disable 2 ^a	PMD transmit disable	1.9.4:3	PMD_transmit_disable_3, PMD_transmit_disable_2
PMD transmit disable 1 ^b	PMD transmit disable	1.9.2	PMD_transmit_disable_1
PMD transmit disable 0	PMD transmit disable	1.9.1	PMD_transmit_disable_0

^a Available only in 400GBASE-CR4.

^b Available only in 200GBASE-CR2 and 400GBASE-CR4.

Table 162–6—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect	1.10.0	Global_PMD_signal_detect
PMD receive signal detect 3 to PMD receive signal detect 2 ^a	PMD receive signal detect	1.10.4:3	PMD_signal_detect_3, PMD_signal_detect_2
PMD receive signal detect 1 ^b	PMD receive signal detect	1.10.2	PMD_signal_detect_1
PMD receive signal detect 0	PMD receive signal detect	1.10.1	PMD_signal_detect_0

^a Available only in 400GBASE-CR4.

^b Available only in 200GBASE-CR2 and 400GBASE-CR4.

Table 162–7—MDIO/PMD control function mapping

MDIO variable	PMA/PMD register name	Register/bit number	PMD variable
Restart training	BASE-R PMD control	1.150.0	mr_restart_training
Training enable	BASE-R PMD control	1.150.1	mr_training_enable
Receiver status	BASE-R PMD status	1.151.0 ^a	local_trained
Frame lock	BASE-R PMD status	1.151.1 ^a	local_tf_lock
Start-up protocol status	BASE-R PMD status	1.151.2 ^a	training
Training failure	BASE-R PMD status	1.151.3 ^a	training_failure
Polynomial identifier	PMD training pattern, lanes 0 to 3	1.1450.12:11 ^b	identifier_i

Table 162–7—MDIO/PMD control function mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/bit number	PMD variable
Seed	PMD training pattern, lanes 0 to 3	1.1450.15:14 1.1450.10:0 ^b	seed_i
Initial condition request	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.13:12 ^b	ic_req
Coefficient select	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.4:2 ^b	coef_sel
Coefficient request	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.1:0 ^b	coef_req
Receiver ready	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.15 ^b	remote_rx_ready
Modulation and precoding status	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.11:10 ^b	remote_tp_mode
Receiver frame lock	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.9 ^b	remote_tf_lock
Modulation and precoding request	BASE-R PAM4 PMD training LD control, lanes 0 to 3	1.1320.11:10 ^b	local_tp_mode
Local receiver ready	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.15 ^b	local_rx_ready
Initial condition status	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.8 ^b	ic_sts
Coefficient select echo	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.5:3 ^b	k
Coefficient status	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.2:0 ^b	coef_sts

^a Bit reference is provided for lane 0. Status for lanes 1 to 3 are located in the same register.

^b Address reference is provided for lane 0. Register for lanes 1 to 3 are located at an offset from the lane 0 register.

162.8 PMD functional specifications

162.8.1 Link block diagram

One direction of a 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 link is shown in Figure 162–2. The PMDs on both ends of the link have connected ground references. Each differential pair in the cable is separately shielded as illustrated in Figure 162–2. The signal shields are connected to ground contacts in the MDI plug connectors on both ends of the cable assembly.

Note that the source lane (SL) signals $SL_i<p>$ and $SL_i<n>$ are the positive and negative sides of the transmitter's differential signal pair on lane i and the destination lane (DL) signals $DL_i<p>$ and $DL_i<n>$ are the positive and negative sides of the receiver's differential signal pair on lane i .

For purposes of system conformance, the PMD sublayer is standardized at the test points described in this subclause.

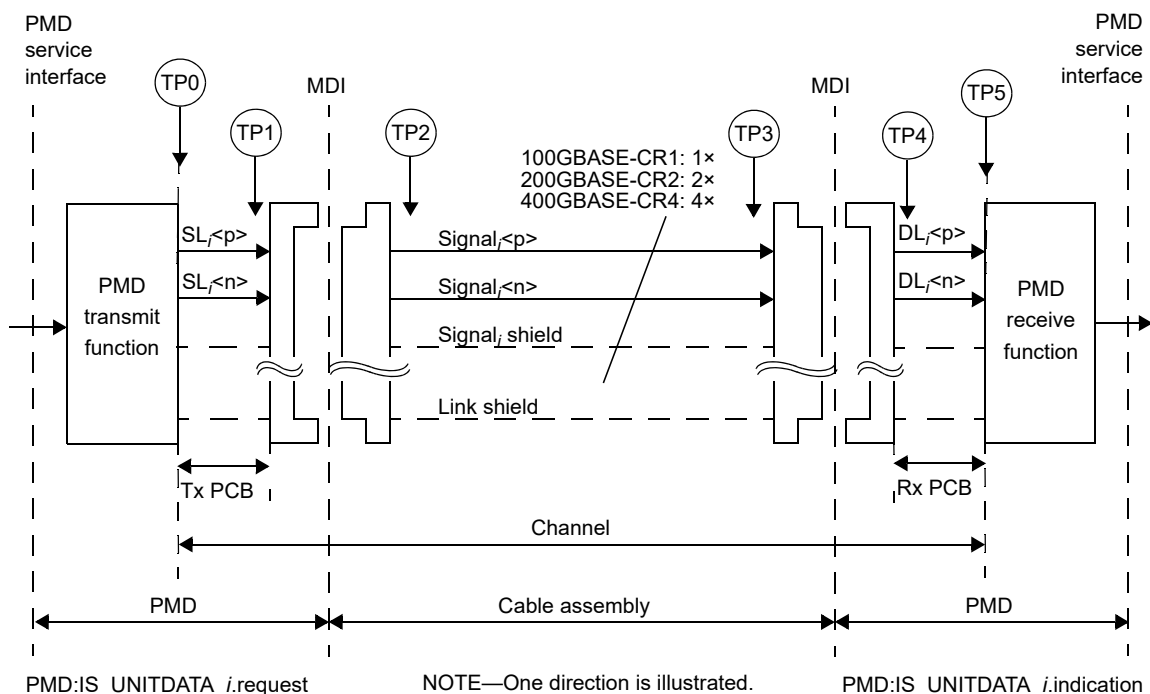


Figure 162-2—100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 link

The electrical transmit signal is defined at TP2. Unless specified otherwise, all transmitter measurements and tests defined in 162.9.4 are made at TP2 utilizing the test fixture specified in Annex 162B.

The electrical receive signal is defined at TP3. Unless specified otherwise, all receiver measurements and tests defined in 162.9.5 are performed at TP3 utilizing the test fixture specified in Annex 162B.

A mated connector pair has been included in both the transmitter and receiver specifications defined in 162.9.4 and 162.9.5. The recommended maximum insertion loss from TP0 to TP2 or TP3 to TP5 including the test fixture is provided in 162.9.4.5. Annex 162A provides information on parameters associated with test points TP0 and TP5, which might not be testable in an implemented system.

The channel (see 162.11) is defined between the transmitter (TP0) and receiver (TP5) blocks to include the transmitter and receiver differential controlled impedance printed circuit boards (PCBs) and the cable assembly, as illustrated in Figure 162-2. All cable assembly measurements are to be made between TP1 and TP4 as illustrated in Figure 162-2. The cable assembly test fixture of Annex 162B.1, or its equivalent, is required for measuring the cable assembly specifications in 162.11 at TP1 and TP4. Two cable assembly test fixtures have been included in the cable assembly specifications defined in 162.11. Transmitter and receiver differential controlled impedance PCB insertion losses defined between TP0 and the MDI and between the MDI and TP5, respectively, are provided informatively in 162A.4.

Table 162–8 describes the defined test points illustrated in Figure 162–2.

Table 162–8—Test points

Test points	Description
TP0 to TP5	The channel including the transmitter and receiver differential controlled impedance PCB insertion loss and the cable assembly insertion loss.
TP1 to TP4	All cable assembly measurements are made between TP1 and TP4 as illustrated in Figure 162–2. The cable assembly test fixture of Annex 162B, or its equivalent, is required for measuring the cable assembly specifications in 162.10 at TP1 and TP4.
TP0 to TP2 TP3 to TP5	A mated connector pair has been included in both the transmitter and receiver specifications defined in 162.9.4 and 162.9.5. The recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture is provided in 162.9.4.5.
TP2	Unless specified otherwise, all transmitter measurements defined in 162.9.4 are made at TP2 utilizing the test fixture specified in Annex 162B.
TP3	Unless specified otherwise, all receiver measurements and tests defined in 162.9.5 are made at TP3 utilizing the test fixture specified in Annex 162B.

162.8.2 PMD transmit function

The PMD transmit function has three operating modes: DATA, TRAINING, and QUIET. The operating mode is controlled by the PMD control state diagram (Figure 136–7).

When operating in DATA mode, the PMD transmit function shall convert the symbol stream requested by the PMD service interface message `PMD:IS_UNITDATA_i.request(tx_symbol)` of each lane into an electrical signal, and deliver the electrical signals to the MDI, according to the transmit electrical specifications in 162.9.4. The differential output voltage ($SL_{i<p>} - SL_{i<n>}$) meets the specifications in 162.9.4.1.1 where the PAM4 symbol values 0, 1, 2, and 3 correspond to the `tx_symbol` values zero, one, two, and three, respectively, with the highest differential output voltage corresponding to `tx_symbol` = three and the lowest differential output voltage corresponding to `tx_symbol` = zero.

When operating in TRAINING mode, the PMD transmit function shall convert the symbol stream generated by the PMD control function of each lane into an electrical signal, and deliver the electrical signals to the MDI, according to the transmit electrical specifications in 162.9.4. The differential output voltage ($SL_{i<p>} - SL_{i<n>}$) meets the specifications in 162.9.4.1.1, with the highest differential output voltage corresponding to the PAM4 symbol 3 and the lowest differential output voltage corresponding to the PAM4 symbol 0.

When operating in QUIET mode the PMD transmit function shall disable the transmitter such that the transmitter drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.

162.8.3 PMD receive function

The PMD receive function shall convert the electrical signal from each of the lanes on the MDI into a symbol stream for delivery to the PMD service interface using the messages `PMD:IS_UNITDATA_i.indication(rx_symbol)`. The highest differential input voltage ($DL_{i<p>} - DL_{i<n>}$) shall correspond to `rx_symbol` = three and the lowest differential input voltage shall correspond to `rx_symbol` = zero.

162.8.4 PMD global signal detect function

For the 100GBASE-CR1 PMD, the variable `Global_PMD_signal_detect` is set to `PMD_signal_detect_0` (see 162.8.5). For the 200GBASE-CR2 and 400GBASE-CR4 PMDs, the variable `Global_PMD_signal_detect` is set to the logical AND of the values of `PMD_signal_detect_i` from each of the PMD lanes.

162.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate the successful completion of the start-up protocol by the PMD control function (see 162.8.11). `PMD_signal_detect_i` shall be set to zero when `signal_detect` on lane *i* is set to false, and to one when `signal_detect` on lane *i* is set to true. The `signal_detect` variables are set independently on each lane by the PMD control state diagram (Figure 136–7).

If training is disabled by the management variable `mr_training_enable` (see 162.7), `PMD_signal_detect_i` shall be set to one for all lanes.

When the MDIO is implemented, this function maps the variables to registers and bits as defined in 162.7.

162.8.6 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional. When implemented, it allows the transmitters on all lanes to be disabled with a single variable.

When `Global_PMD_transmit_disable` variable is set to one, this function shall turn off all the transmitters such that each transmitter drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.

If a PMD fault (162.8.8) is detected, then the PMD may set `Global_PMD_transmit_disable` to one.

162.8.7 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the transmitter in each lane to be selectively disabled. When this function is supported, it shall meet the following requirements:

- When a `PMD_transmit_disable_i` variable is set to one, this function shall turn off the transmitter in lane *i* so that it drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.
- If a PMD fault (162.8.8) is detected, then the PMD may set each `PMD_transmit_disable_i` to one, turning off the transmitter in each lane.

162.8.8 PMD fault function

`PMD_fault` is the logical-OR of `PMD_receive_fault`, `PMD_transmit_fault`, and any other implementation specific fault. If the MDIO interface is implemented, then `PMD_fault` shall be mapped to the fault bit as specified in 45.2.1.2.3.

162.8.9 PMD transmit fault function (optional)

The PMD transmit fault function is optional. The faults detected by this function are implementation specific, but the assertion of `Global_PMD_transmit_disable` is not considered a transmit fault.

If `PMD_transmit_fault` is set to one, then `Global_PMD_transmit_disable` should also be set to one.

If the MDIO interface is implemented, then PMD_transmit_fault shall be mapped to the Transmit fault bit as specified in 45.2.1.7.4.

162.8.10 PMD receive fault function (optional)

The PMD receive fault function is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable PMD_receive_fault to one.

If the MDIO interface is implemented, then PMD_receive_fault shall be mapped to the Receive fault bit as specified in 45.2.1.7.5.

162.8.11 PMD control function

The PMD control function performs the PMD start-up protocol. This protocol facilitates timing recovery and equalization while providing a mechanism through which the receiver can configure the transmitter to optimize performance. The protocol supports these functions through the continuous exchange of fixed-length training frames.

The PMD shall implement one instance of the PMD control function described in 136.8.11 for each lane with the following exceptions:

- a) The control field structure is specified in Table 162–9 and the status field structure is specified in Table 162–10.
- b) For k_list as specified in 136.8.11.4.4, the set of valid transmitter equalizer coefficient indices is $\{-3, -2, -1, 0, +1\}$.
- c) For the initial condition request as described in 136.8.11.2.1, five predefined transmitter equalizer settings are specified in 162.9.4.1.3.
- d) The “No equalization” value (see 136.8.11.2.4) of $c(-3)$ is 0.
- e) The terminal count of max_wait_timer as specified in 136.8.11.7.3 is 12 s.
- f) A receiver is expected to assert local_tf_lock within 275 ms from entry into the AN_GOOD_CHECK state in Figure 73–11 provided that there is a compliant signal containing valid training frames at the PMD input.
- g) The value of use_quiet_in_training (see 136.8.11.7.1) is TRUE.

Table 162–9—Control field structure

Bit(s)	Name	Description
15:14	Reserved	Transmit as 0, ignore on receipt
13:11	Initial condition request	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control
10	Reserved	Transmit as 0, ignore on receipt
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2
7:5	Reserved	Transmit as 0, ignore on receipt
4:2	Coefficient select	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved
1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold

Table 162–10—Status field structure

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14:12	Reserved	Transmit as 0, ignore on receipt
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Initial condition status	1 = Updated 0 = Not updated
7	Parity	Even parity bit
6	Reserved	Transmit as 0, ignore on receipt
5:3	Coefficient select echo	5 4 3 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$
2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated

162.9 PMD electrical characteristics

162.9.1 AC-coupling

Interoperability between PMD components operating from different supply voltages is facilitated by AC-coupling within the cable assembly as specified in 162.11.

162.9.2 Signal paths

The MDI transmit and receive paths are point-to-point connections. Each path corresponds to one MDI lane and comprises two complementary signals, which form a balanced differential pair.

For 100GBASE-CR1, there is one differential lane in each direction for a total of two pairs, or four connections. For 200GBASE-CR2, there are two differential lanes in each direction for a total of four pairs, or eight connections. For 400GBASE-CR4, there are four differential lanes in each direction for a total of eight pairs, or sixteen connections.

162.9.3 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

162.9.4 Transmitter characteristics

The transmitter on each lane shall meet the specifications given in Table 162–11 and detailed in the referenced subclauses. Unless specified otherwise, all transmitter signal measurements are made for each lane separately, at TP2, utilizing the test fixtures specified in Annex 162B and a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth. The connection from TP2 to the test equipment is AC-coupled.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP2 where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

Recommended transmitter characteristics at TP0 are provided in 162A.2.

Table 162–11—Summary of transmitter specifications at TP2

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)		53.125 $\pm$ 50 ppm ^a	GBd
Differential pk-pk voltage with Tx disabled (max) ^b	93.8.1.3	30	mV
DC common-mode voltage (max) ^b	93.8.1.3	1.9	V
AC common-mode peak-to-peak voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	162.9.4.4	30 80	mV mV
Differential pk-pk voltage, v_{di} (max) ^b	93.8.1.3	1200	mV
Effective return loss, ERL (min)	162.9.4.8	7.3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	162.9.4.9	See Equation (162–6)	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	162.9.4.10	See Equation (162–7)	dB
Transmitter steady-state voltage, v_f (min) Transmitter steady-state voltage, v_f (max)	162.9.4.1.2	0.387 0.6	V
Linear fit pulse peak ratio, R_{peak} (min)	162.9.4.1.2	0.397	—
Level separation mismatch ratio R_{LM} (min)	162.9.4.2	0.95	—
Transmitter output waveform absolute value of step size for all taps (min) absolute value of step size for all taps (max) value at minimum state for $c(-3)$ (max) value at maximum state for $c(-2)$ (min) value at minimum state for $c(-1)$ (max) value at minimum state for $c(0)$ (max) value at minimum state for $c(1)$ (max)	162.9.4.1.4 162.9.4.1.4 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5	0.005 0.025 –0.06 0.12 –0.34 0.5 –0.2	— — — — — — —

Table 162–11—Summary of transmitter specifications at TP2 (*continued*)

Parameter	Subclause reference	Value	Units
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	31.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	162.9.4.3	26.7	dB
Output jitter (max)			
J_{RMS}	162.9.4.7	0.023	UI
J_{3u03}	162.9.4.7	0.115	UI
J_{3u}	162.9.4.7	0.125	UI
Even-odd jitter, pk-pk	162.9.4.7	0.025	UI

^a For a PMD in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

162.9.4.1 Transmitter output waveform

The transmit function includes programmable equalization to compensate for the frequency-dependent loss of the channel and facilitate data recovery at the receiver. The functional model for the transmit equalizer on each lane is the five-tap transversal filter shown in Figure 162–3.

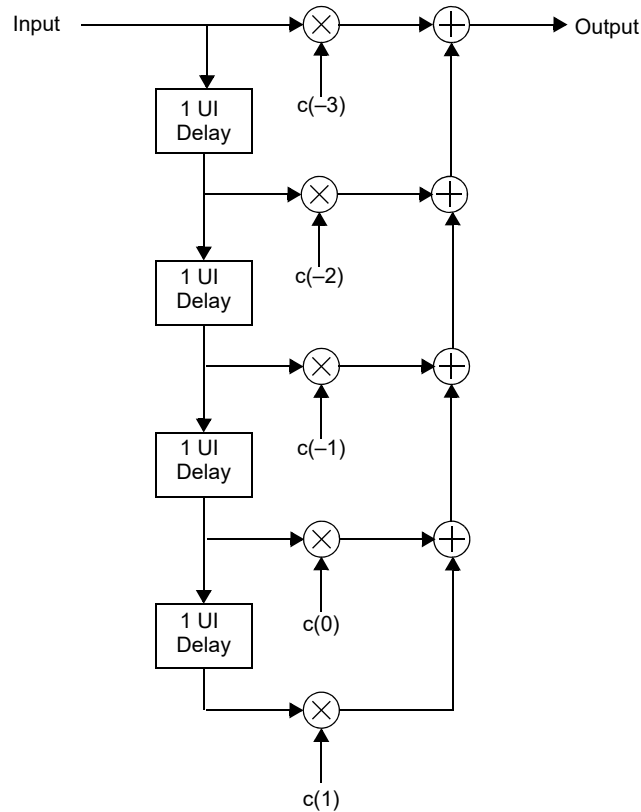


Figure 162–3—Transmit equalizer functional model

The state of the transmit equalizer and hence the transmitted output waveform may be manipulated via the PMD control function defined in 162.8.11 or via the management interface. The transmit function responds

to a set of commands issued by the link partner's receive function and conveyed by a back-channel communications path. This command set includes instructions to increment, decrement, hold, or set to zero a selected coefficient $c(k)$, where $k = -3$ to 1. In addition, it includes commands to set all coefficients to one of five initial conditions.

In response, the transmit function relays status information to the link partner's receive function. The status messages indicate the selected coefficient index k , the status of the coefficient $c(k)$ (updated, not updated, or at limit), and the status of the initial condition (updated or not updated).

162.9.4.1.1 Linear fit to the measured waveform

The following procedure is used to determine the linear fit pulse response, linear fit error, and normalized transmitter coefficient values.

Set the transmitter under test to transmit the PRBS13Q test pattern (defined in 120.5.11.2.1). For each configuration of the transmit equalizer, capture at least one complete cycle of the test pattern at TP2, as specified in 85.8.3.3.4. The reference clock recovery unit (CRU) used in the measurement acts as a high-pass jitter filter with a corner frequency of 4 MHz and a slope of 20 dB/decade.

In the following calculations, M should be an integer greater than or equal to 32. Interpolation of the captured waveform may be used to achieve this.

Compute the linear fit pulse response $p(k)$ and linear fit error $e(k)$, $k=1$ to $M \times N_p$, from the captured waveform, as specified in 85.8.3.3.5, with $N_p = 200$ and $D_p = 4$, where the aligned symbols $x(n)$ are assigned normalized amplitudes -1 , $-ES$, ES , and 1 to represent the PAM4 symbol values 0, 1, 2, and 3 respectively. ES is defined as $(|ES1| + |ES2|)/2$ where $ES1$ and $ES2$ are calculated according to 120D.3.1.2.

Define $r(k)$ to be the linear fit pulse response when transmit equalizer coefficients have been set to the "preset 1" values. The normalized coefficients for any configuration of the transmit equalizer are computed using the following method.

For each value of m in the range $-M/2$ to $M/2 - 1$ when M is even and $-(M-1)/2$ to $(M-1)/2$ when M is odd:

- a) Define an $M \times N_p$ -by-5 matrix R_m . The elements of R_m are assigned values per Equation (162-1) where $i = -3$ to 1 and $j = 1$ to $M \times N_p$.

$$R_m(j, i+4) = \begin{cases} r(m+j-M \times i) & \text{if } 1 \leq m+j-M \times i \leq M \times N_p \\ 0 & \text{otherwise} \end{cases} \quad (162-1)$$

- b) The normalized coefficients of the transmit equalizer are computed using Equation (162-2).

$$\begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} = (R_m^T R_m)^{-1} R_m^T \begin{bmatrix} p(1) \\ \dots \\ p(M \times N_p) \end{bmatrix} \quad (162-2)$$

- c) The linear fit pulse response is reconstructed from the matrix R_m and the normalized coefficients using Equation (162-3).

$$\begin{bmatrix} p_m(1) \\ \dots \\ p_m(M \times N_p) \end{bmatrix} = R_m \begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} \quad (162-3)$$

d) $\varepsilon^2(m)$ is computed using Equation (162-4).

$$\varepsilon^2(m) = \sum_{k=1}^{M \times N_p} (p(k) - p_m(k))^2 \quad (162-4)$$

The normalized transmit equalizer coefficients $c(i)$ for a given linear fit pulse $p(k)$ are the values $c_m(i)$ for the value of m that minimizes $\varepsilon^2(m)$.

162.9.4.1.2 Steady-state voltage and linear fit pulse peak ratio

The linear fit pulse peak v_{peak} and steady-state voltage v_f are defined using the linear fit pulse response $p(1)$ through $p(M \times N_p)$, measured with transmit equalizer set to preset 1 (no equalization). N_p is set equal to 200. The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 162.9.4.1.1.

v_{peak} is defined as the maximum value of $p(k)$.

v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M .

The linear fit pulse peak ratio R_{peak} is defined by Equation (162-5).

$$R_{peak} = \frac{v_{peak}}{v_f} \quad (162-5)$$

162.9.4.1.3 Coefficient initialization

When the Coefficient update state diagram (Figure 136-9) is in either the OUT_OF_SYNC state or the NEW_IC state, the coefficients of the transmit equalizer shall be configured to values within the ranges specified in Table 162-12 (according to the value of the variable `ic_req`, which is defined in 136.8.11.7.1). These requirements apply upon the assertion of initial condition status of “Updated”.

162.9.4.1.4 Coefficient step size

The change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “increment” shall be between 0.005 and 0.025, and the change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “decrement” shall be between −0.025 and −0.005. It is recommended that the same step size is used for all coefficients.

The coefficients other than $c(\text{coef_sel})$ are not expected to change. The absolute change in any coefficient other than $c(\text{coef_sel})$ shall be less than 0.005.

The change in the normalized transmit equalizer coefficient is defined to be the difference in the value measured prior to the assertion of the “increment” or “decrement” request (i.e., `coef_req` is “Hold”) and the value upon the assertion of a coefficient status of “updated”.

Table 162–12—Coefficient initial conditions

Coefficient update state	ic_req	$c(-3)$	$c(-2)$	$c(-1)$	$c(0)$	$c(1)$
OUT_OF_SYNC ^a	N/A	0	0	0	1	0
NEW_IC	preset 1 ^a	0	0	0	1	0
	preset 2	0 ± 0.0125	0 ± 0.0125	0 ± 0.0125	0.5 ± 0.0125	0 ± 0.0125
	preset 3	0 ± 0.0125	0 ± 0.0125	-0.075 ± 0.0125	0.75 ± 0.0125	0 ± 0.0125
	preset 4	0 ± 0.0125	0.05 ± 0.0125	-0.2 ± 0.0125	0.75 ± 0.0125	0 ± 0.0125
	preset 5	-0.025 ± 0.0125	0.075 ± 0.0125	-0.25 ± 0.0125	0.65 ± 0.0125	0 ± 0.0125

^a PRESET1 is the reference for the calculation of the normalized coefficients of the transmit equalizer (see 162.9.4.1.1). As a result the normalized coefficients for PRESET1 and OUT_OF_SYNC do not include any tolerances.

162.9.4.1.5 Coefficient range

When sufficient “increment” or “decrement” requests have been received for a given coefficient, the coefficient reaches a lower or upper bound based on the range of that coefficient or the combination of coefficients.

With $c(-3)$, $c(-2)$, and $c(-1)$ set to zero and both $c(0)$ and $c(1)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(1)$ shall be less than or equal to -0.2.

With $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$ set to zero and having received sufficient “decrement” requests so that it is at its minimum value, $c(0)$ shall be less than or equal to 0.5.

With $c(-3)$, $c(-2)$, and $c(1)$ set to zero and both $c(-1)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-1)$ shall be less than or equal to -0.34.

With $c(-3)$, $c(-1)$, and $c(1)$ set to zero, $c(0)$ having received sufficient “decrement” requests so that it is at its minimum value, and $c(-2)$ having received sufficient “increment” requests so that it is at its maximum value, $c(-2)$ shall be greater than or equal to 0.12.

With $c(-2)$, $c(-1)$, and $c(1)$ set to zero and both $c(-3)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-3)$ shall be less than or equal to -0.06.

NOTE—Any of the coefficients $c(-3)$, $c(-2)$, $c(-1)$, or $c(1)$ may be set to zero by asserting a coefficient request of “no equalization” for that coefficient, using the control function specified in 162.8.11, or by implementation specific means.

162.9.4.2 Transmitter linearity

Transmitter linearity is defined using the method in 120D.3.1.2.

The transmitter linearity shall meet the requirement when the transmitter equalization is set to each of the initial conditions defined in Table 162–12.

162.9.4.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is determined using the method in 120D.3.1.7 with the following exceptions:

- The linear fit pulse response $p(k)$ is determined using the linear fit procedure in 162.9.4.1.1.
- The continuous time filter parameters are provided in Table 162–20.
- For calculation of SNR_{ISI} using Equation (120D–8) a value of 6 is used for N_b .
- A time offset is added to t_p whose value is swept from -0.5 UI to 0.5 UI when calculating $ISI_{cursors}$. SNR_{ISI} is defined as the lowest value found across the time offset sweep.
- The transmitter equalizer setting is chosen, within the required coefficient ranges (see 162.9.4.1.5), to result in the highest SNR_{ISI} value.

162.9.4.4 Peak-to-peak AC common-mode voltage

Full-band peak-to-peak AC common-mode voltage, $VC_{M_{FB}}$, is defined as the AC common-mode voltage (see 93.8.1.3) range that includes all but 10^{-4} of the measured distribution, from 0.00005 to 0.99995 of the cumulative distribution. The transmitter equalization is turned off (preset 1 condition).

Low-frequency peak-to-peak AC common-mode voltage, $VC_{M_{LF}}$, is defined in the same way as $VC_{M_{FB}}$, except that it is measured with a low-pass filter defined by Equation (93A–20) with f_c set to 100 MHz.

The low-frequency and full-band peak-to-peak AC common-mode voltages shall meet the specifications for $VC_{M_{LF}}$ (max) and $VC_{M_{FB}}$ (max) in Table 162–11.

NOTE 1—The common-mode noise measurement may be sensitive to mismatches between the single-ended paths in the test fixture and the test setup. Careful design and calibration of the test system are recommended.

NOTE 2—The common-mode noise measurement should take into consideration frequencies down to the specified AC-coupling frequency limit (e.g., the 50 kHz limit specified in 162.11 for the cable assembly).

162.9.4.5 Insertion loss, TP0 to TP2 or TP3 to TP5

The recommended insertion loss for TP0 to TP2 and for TP3 to TP5 is provided in 162A.4.

162.9.4.6 Output SNDR

The transmitter SNDR is defined by the measurement method described in 120D.3.1.6, with the exceptions that the test system response is specified in 162.9.4 and the linear fit procedure in 162.9.4.1.1 is used.

The transmitter SNDR shall meet the requirement when the transmitter equalization is set to each of the initial conditions defined in Table 162–12.

162.9.4.7 Output jitter

Output jitter is characterized by four parameters: J_{RMS} , J_{3u} , $J_{3u_{03}}$, and even-odd jitter. These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the transmitter package and host channel. The equalizer setting is chosen to minimize any or all of the jitter parameters.

J_{RMS} is calculated using the measurement method specified in 120D.3.1.8.1. J_{3u} is calculated using the measurement method for J_{4u} in 120D.3.1.8.1, except that J_{3u} is defined as the time interval that includes all but 10^{-3} of $f_j(t)$, from the 0.05th to the 99.95th percentile of $f_j(t)$. $J_{3u_{03}}$ is calculated the same way as J_{3u} except that the jitter calculation uses only transitions R03 and F30 in Table 162–13.

Even-odd jitter is calculated using the measurement method specified in 120D.3.1.8.2. with the following exceptions:

- The test pattern is either PRBS13Q or alternatively PRBS9Q. PRBS9Q is defined in 120.5.11.2.a. Meeting the even-odd jitter requirement with only one pattern is sufficient.
- If the test pattern is PRBS13Q, the corner frequency of the clock recovery unit (CRU) is set to 4 MHz or to 1 MHz. Meeting the even-odd jitter requirement with only one CRU bandwidth is sufficient.

NOTE 1—If the measuring instrument is triggered by a clock based on the signaling rate divided by an even number, the even-odd jitter might not be correctly observed. As a result, the observation of J_{3u} and J_{RMS} might also be affected.

NOTE 2— J_{3u} is sensitive to measurement noise being converted to timing errors. Hence, accounting for measurement noise effects is recommended.

The pattern symbols and thresholds used to define each transition for PRBS13Q are defined in Table 120D-4. The pattern symbols and thresholds used to define each transition for PRBS9Q are defined in Table 162-13.

Table 162-13—PRBS9Q pattern symbols used for jitter measurement

Label	Description	Gray coded PAM4 symbols	Index of first symbol	Index transition begins	Index transition ends	Index of last symbol	Threshold level
REF	Reference for symbol index	33333	1	—	—	5	—
R03	0 to 3 rise	1000 331	260	263	264	266	$(V_0+V_3)/2^a$
F30	3 to 0 fall	233333 001	511	5	6	8	
R12	1 to 2 rise	311 2221	464	466	467	470	$(V_1+V_2)/2$
F21	2 to 1 fall	122 11110	254	256	257	261	
R01	0 to 1 rise	200 113	503	505	506	508	$(V_0+V_1)/2$
F10	1 to 0 fall	21111 0003	256	260	261	264	
R23	2 to 3 rise	3222 330	210	213	214	216	$(V_2+V_3)/2$
F32	3 to 2 fall	133 223	507	509	510	1	
R02	0 to 2 rise	200 22223	63	65	66	70	$(V_0+V_2)/2$
F20	2 to 0 fall	12222 001	321	325	326	328	
R13	1 to 3 rise	0111 331	166	169	170	172	$(V_1+V_3)/2$
F31	3 to 1 fall	033 1112	263	265	266	269	

^a V_0 , V_1 , V_2 , and V_3 are defined in 120D.3.1.2.

162.9.4.8 Transmitter effective return loss (ERL)

The ERL of the transmitter at TP2 is defined by the procedure in 93A.5 using the values in Table 162–14 and Table 162–20, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Table 162–14—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

Transmitter ERL at TP2 shall meet the requirement in Table 162–11.

162.9.4.9 Common-mode to common-mode return loss

The transmitter common-mode to common-mode return loss shall meet Equation (162–6).

$$RL_{cc}(f) \geq \begin{cases} 2 & 0.2 \leq f < 4 \\ 1.6 + 0.1f & 4 \leq f \leq 30 \\ 8.5 - 0.13f & 30 < f \leq 40 \end{cases} \quad (162-6)$$

where

$RL_{cc}(f)$ is the transmitter common-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

The transmitter common-mode to common-mode return loss limit is illustrated in Figure 162–4.

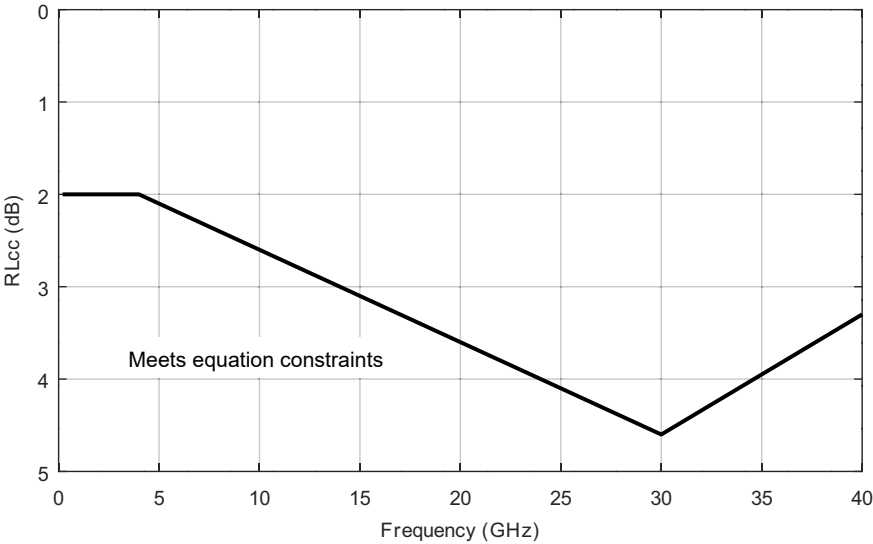


Figure 162-4—Transmitter common-mode to common-mode return loss

162.9.4.10 Common-mode to differential-mode return loss

The transmitter common-mode to differential-mode return loss shall meet Equation (162-7).

$$RL_{dc}(f) \geq \left\{ \begin{array}{ll} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{array} \right\} \quad (162-7)$$

where

$RL_{dc}(f)$ is the transmitter common-mode to differential-mode return loss at frequency f in dB
 f is the frequency in GHz

The transmitter common-mode to differential-mode return loss limit is illustrated in Figure 162-5.

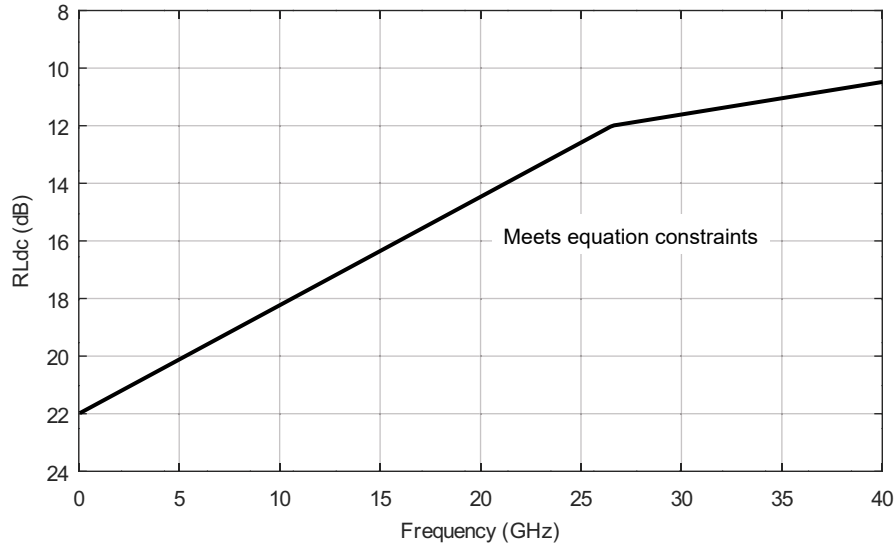


Figure 162-5—Transmitter common-mode to differential-mode return loss

162.9.5 Receiver characteristics

The receiver on each lane shall meet the specifications given in Table 162-15 and detailed in the referenced subclauses. Unless specified otherwise, all receiver measurements are made for each lane separately, at TP3, utilizing the test fixtures specified in Annex 162B.

Recommended receiver characteristics at TP5 are provided in 162A.3.

Table 162-15—Summary of receiver specifications at TP3

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)	162.9.5.1	53.125 ± 100 ppm	GBd
Amplitude tolerance	162.9.5.2	1200 ^a	mV
Interference tolerance	162.9.5.3	Table 162-16	—
Jitter tolerance	162.9.5.4	Table 162-17	—
Effective return loss, ERL (min)	162.9.5.5	7.3	dB
Differential-mode to common-mode return loss, <i>RL_{cd}</i> (min)	162.9.5.6	Equation (162-18)	dB

^a Amplitude is measured at TP2.

162.9.5.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 162.9.5.2, 162.9.5.3, and 162.9.5.4 for any signaling rate in the range 53.125 GBd ± 100 ppm.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ± 50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ± 100 ppm, when derived from an intermediate interface (e.g., 100GAUI-4).

162.9.5.2 Receiver amplitude tolerance

When a PMD receiver is connected to a compliant transmitter whose peak-to-peak differential output voltage (see Table 162–11 footnote b) measured at the preset 1 equalizer setting is 1200 mV, using a compliant cable assembly with the minimum insertion loss specified in 162.11.2, the PMD receiver operation shall enable an FEC symbol error ratio better than 10^{-3} assuming errors are sufficiently uncorrelated to support the required frame loss ratio (see 162.1). If errors are not sufficiently uncorrelated, the FEC symbol error ratio shall be lower as appropriate to support the required frame loss ratio.

The receiver is allowed to control the transmitter equalizer coefficients, using the PMD control function defined in 162.8.11 or an equivalent process, to meet these requirements.

162.9.5.3 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 162.9.5.3.1 through 162.9.5.3.5. Receiver interference tolerance test requirements are specified in Table 162–16.

Two tests are defined. Test 1 includes a low-loss channel. Test 2 includes a high-loss channel. The cable assembly used in the test channel (see 110.8.4.2.2) shall meet the cable assembly Channel Operating Margin (COM) as specified in 162.11.7.

Table 162–16—Interference tolerance test parameters

Parameter	Test 1 (low loss)		Test 2 (high loss)		Units
	Min	Max	Min	Max	
Test pattern	Scrambled idle encoded by FEC				
FEC symbol error ratio required ^a	$< 10^{-3}$				
Test channel insertion loss at 26.56 GHz ^b	10.5	11.5	23.625	24.625	dB
Cable assembly insertion loss at 26.56 GHz	10.5	11.5	17.75	19.75	dB
COM ^c	3		3		dB

^a See 162.9.5.3.5 for definition of FEC symbol error ratio.

^b Insertion loss between the two test references (see Figure 110–3b).

^c The COM value is the target value for the SNR_{TX} calibration defined in 162.9.5.3.3 item h). The SNR_{TX} value measured at the Tx test reference should be as close as practical to the value needed to produce the target COM. If lower SNR_{TX} values are used, this would demonstrate margin to the specification but this is not required for compliance.

162.9.5.3.1 Test setup

The interference tolerance test is performed with the setup similar to the one described in 110.8.4.2.1. The test setup includes noise injection for all lanes of the pattern generator.

162.9.5.3.2 Test channel

The test channel is the same as the one defined in 110.8.4.2.2, except that the cable assembly meets the requirements of 162.11, the test channel loss meets the requirements of Table 162–16 and the cable assembly test fixture meets the requirements of 162B.3.

162.9.5.3.3 Test channel calibration

The scattering parameters of the test channel are measured at the test references as illustrated in Figure 110–3b using the cable assembly test fixtures specified in Annex 162B.1.

The insertion loss at 26.56 GHz of the signal path between the test references in Figure 110–3b is within the limits in Table 162–16.

The COM is calculated using the method and parameters of 162.11.7 with the following considerations:

- a) The channel signal path is $SCHS_p = \text{cascade}(S^{(CTSP)}, S^{(HOSPR)})$, where $\text{cascade}()$ is defined in 93A.1.2.1, $S^{(HOSPR)}$ is defined in 162.11.7.1.1, and $S^{(CTSP)}$ is the measured channel between the test references in Figure 110–3b.
- b) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 162–20 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- c) The augmented signal path in 93A.1.2 is replaced by S_p determined from Equation (162–8) (effectively omitting the transmitter device package model $S^{(tp)}$). The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (93A–19) uses T_r equal to the transition time at the Tx test reference. T_r is measured using the method in 120E.3.1.5 with the transmit equalizer turned off (i.e., coefficients set to the preset 1 values, see 162.9.4.1.3) with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 40 GHz.
- d) Even-odd jitter, J3u, and J_{RMS} without noise injection (see 162.9.5.3.4) are measured at the Tx test reference and comply with the specification in Table 162–11. In the calculation of COM, A_{DD} , and σ_{RJ} are calculated from the measured values of J3u and J_{RMS} using Equation (162–9) and Equation (162–10), replacing the values in Table 162–20. It is recommended to adjust the pattern generator jitter such that J3u and J_{RMS} are as close as practical to their limits in Table 162–11.
- e) In the calculation of COM, SNR_{TX} is set to the SNDR value measured at the Tx test reference using the procedure in 120D.3.1 with the exception that the linear fit in 120D.3.1.3 is performed with a pulse length (N_p) of 29 UI and with pattern generator noise disabled.
- f) For the CTLE and FFE settings used in the calculation of COM in items a) to e), find the value of σ_{TX} required to achieve the COM value specified in Table 162–16. From σ_{TX} , determine the value of σ_{ne} using Equation (162–11).
- g) From σ_{ne} , the voltage transfer function $H_{21}^{(0)}(f)$ of the test channel, the transfer function $H_{ctf}(f)$ of the CTLE used in COM, and $H_{hp}(f)$ defined in Equation (162–13), find σ_{bn} , using Equation (162–12). From σ_{bn} and $H_{hp}(f)$ find σ_{hp} using Equation (162–14).
- h) Set the RMS voltage of the noise added to the signal before the Tx test reference to σ_{hp} .

$$S_p = \text{cascade}(SCHS_p, S^{(rp)}) \quad (162-8)$$

where

$\text{cascade}()$	is defined in 93A.1.2.1
$SCHS_p$	is defined in item a) above
$S^{(rp)}$	is defined in 93A.1.2.4

$$A_{DD} = \frac{\frac{J3u}{2} + Q3d \sqrt{(Q3d^2 + 1) \times J_{RMS}^2 - \left(\frac{J3u}{2}\right)^2}}{Q3d^2 + 1} \quad (162-9)$$

$$\sigma_{RJ} = \frac{\frac{J3u}{2} - A_{DD}}{Q3d} \quad (162-10)$$

where

$$Q3d = 3.0902$$

NOTE 1— $Q3d$ is an approximated solution of $Q(Q3d) = 10^{-3}$, where the Q function is defined in Equation (95-1).

NOTE 2—The calculation of A_{DD} may, under certain conditions, pose a negative discriminant. If this condition occurs, the recommended solution is to increase sinusoidal jitter or bounded uncorrelated jitter such that the discriminant is positive.

$$\sigma_{TX}^2 = [h^{(0)}(t_s)]^2 10^{-SNR_{TX}/10} + \sigma_{ne}^2 \quad (162-11)$$

$$\sigma_{ne}^2 = \frac{\sigma_{bn}^2}{(f_b/2)} \int_0^{f_b/2} |H_{hp}(f)H_{21}^{(0)}(f)H_r(f)H_{ctf}(f)|^2 df \quad (162-12)$$

$$H_{hp}(f) = \frac{j \frac{f}{f_{hp}}}{1 + j \frac{f}{f_{hp}}} \quad (162-13)$$

$$\sigma_{hp}^2 = \frac{\sigma_{bn}^2}{(f_b/2)} \left(\int_0^{f_b/2} |H_{hp}(f)|^2 df \right) \approx 0.6954 \sigma_{bn}^2 \quad (162-14)$$

where

σ_{TX}	is the RMS transmitter noise
SNR_{TX}	is the transmitter signal-to-noise ratio
$H_{hp}(f)$	is the high-pass spectral shaping of the noise
f_{hp}	is the 3 dB cutoff frequency of the high-pass spectral shaping filter with a value of 6 GHz
σ_{bn}	is the RMS broadband noise amplitude
σ_{ne}	is the RMS filtered noise amplitude
σ_{hp}	is the RMS noise amplitude added at the transmitter test reference
$h^{(0)}$	is defined in 93A.1.5.
$H_{21}^{(0)}(f)$	is defined in 93A.1.3
$H_r(f)$	is defined in 93A.1.4.1
$H_{ctf}(f)$	is defined in 93A.1.4.3

162.9.5.3.4 Pattern generator and noise injection

The pattern generator transmits data to all lanes of the device under test. At the start of transmitter training, the pattern generator output amplitude on all lanes shall be 800 mV peak-to-peak differential when measured on an alternating zero-three sequence. The output amplitude, measured on an alternating zero-three sequence, is not permitted to exceed 800 mV peak-to-peak differential during transmitter training. The output waveform and the ERL of the pattern generator comply with 162.9.4.

Noise is added to the signal before the Tx test reference, with noise level set according to step h) in 162.9.5.3.3.

The noise required for each lane is calibrated. The noise may be added either to one lane at a time or using multiple noise sources to all lanes at the same time.

To avoid excessive low-frequency weighting to the receiver input noise, the noise added to the signal is bounded by the normalized spectral density mask defined in Equation (162–15), Equation (162–16), and Equation (162–17) with $f_1 = 8$ GHz and $f_2 = 5$ GHz, and illustrated in Figure 162–6.

$$NSD_{avg} = \frac{1}{(f_b/2 - f_1)} \int_{f_1}^{f_b/2} NSD(f) df \quad (162-15)$$

$$10\log_{10}\left(\frac{NSD(f)}{NSD_{avg}}\right) \geq -3 + 3.6(f/f_b) \quad f_1 \leq f \leq f_b/2 \quad (162-16)$$

$$10\log_{10}\left(\frac{NSD(f)}{NSD_{avg}}\right) \leq \begin{cases} -12 + 15(f/f_2) & 0 \leq f < f_2 \\ 3 & f_2 \leq f \leq f_b/2 \end{cases} \quad (162-17)$$

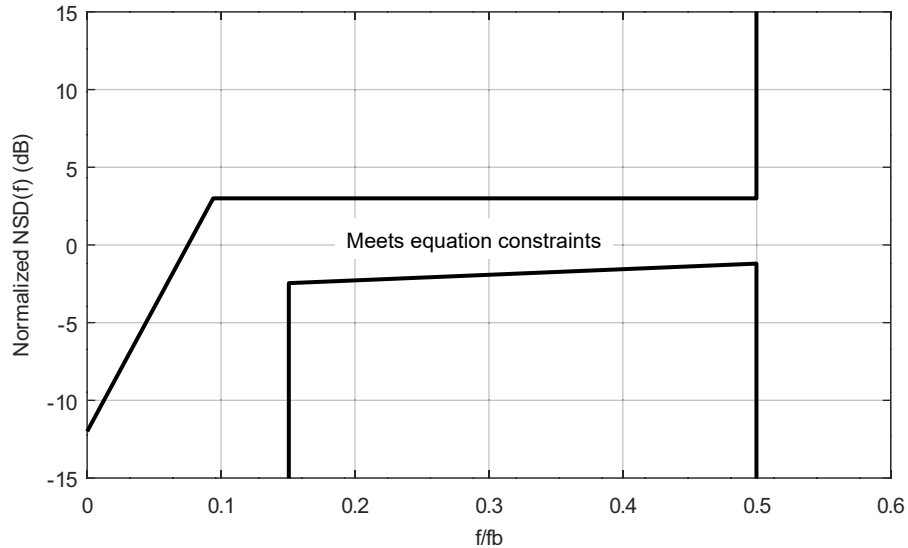


Figure 162–6—NSD(f) constraints

162.9.5.3.5 Test procedure

The pattern generator is first configured to transmit the training pattern defined in 162.8.11. During this initialization period, the device under test (DUT) configures the pattern generator transmit equalizer to the coefficient settings it would select using the protocol described in 162.8.11 and the receiver is tuned using its optimization method. The coefficient settings may be communicated via the start-up protocol or by other means.

After the pattern generator equalizer has been configured and the receiver tuned, the pattern generator is set to generate the test pattern specified in Table 162–16. During the test, the transmitters in the device under test transmit the same pattern type specified for the test, with equalization turned off (preset 1 condition).

Symbol error ratio is measured using the per-lane FEC symbol error counters (see 91.6 or 161.6) in the adjacent FEC sublayer, or the per-lane PCS symbol error counters (see 119.3.1) in the adjacent PCS, as appropriate.

The FEC symbol error ratio is defined as the sum of the symbol error counters on all lanes divided by the number of symbols transmitted on all lanes, which may be estimated from the test time.

A PHY shall meet the FEC symbol error ratio requirement in all tests defined in Table 162–16 with broadband noise added to all lanes (see 162.9.5.3.4). The FEC symbol error ratio requirement assumes that errors are sufficiently uncorrelated to support the required frame loss ratio (see 162.1). If errors are not sufficiently uncorrelated, the FEC symbol error ratio shall be lower as appropriate to support the required frame loss ratio.

NOTE—If noise is applied to each of the n lanes one at a time, results of the n measurements are summed to yield the FEC symbol error ratio. The result may need to be corrected based on the FEC symbol error ratio with no noise added on any lane.

162.9.5.4 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 162.9.5.4.1 and 162.9.5.4.2

162.9.5.4.1 Test setup

Jitter tolerance is measured with a channel meeting the insertion loss of Test 2 as specified in Table 162–16. The pattern generator includes sinusoidal jitter injection.

162.9.5.4.2 Test procedure

The jitter tolerance test procedure is similar to that of 162.9.5.3, with the exception that no noise is injected (i.e., step h) in 162.9.5.3.3 is not performed). Instead, jitter with the specified frequency and amplitude is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency in Table 162–17 at the Tx test reference (see Figure 110–3a). The test channel COM, calculated per 162.9.5.3.3 with the jitter-stressed transmitter output, shall not be lower than the value in Table 162–16.

A PHY shall meet the FEC symbol error ratio requirement defined in Table 162–16 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17 with jitter added to all lanes.

NOTE 1—If jitter is applied to each of the n lanes one at a time, results of the n measurements are summed to yield the FEC symbol error ratio. The result may need to be corrected based on the FEC symbol error ratio with no jitter applied on any lane.

NOTE 2—The A_{DD} (Equation (162–9)) and σ_{RJ} (Equation (162–10)) calculated from transmitter measurements in this test may be higher than the values in Table 162–20. A suitable channel should be chosen in order to meet the COM requirement with these values.

Table 162–17—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Units
Jitter frequency	0.04	0.4	1.333	4	12	40	MHz
Jitter amplitude (pk-pk)	5	0.5	0.15	0.05	0.05	0.05	UI

162.9.5.5 Receiver ERL

The ERL of the receiver at TP3 is computed using the procedure in 93A.5 with the values in Table 162–14 and Table 162–20, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Receiver ERL at TP3 shall meet the requirement specified in Table 162–15.

162.9.5.6 Differential-mode to common-mode return loss

The receiver differential-mode to common-mode return loss shall meet Equation (162–18) as illustrated in Figure 162–7.

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (162-18)$$

where

$RLcd(f)$ is the receiver differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

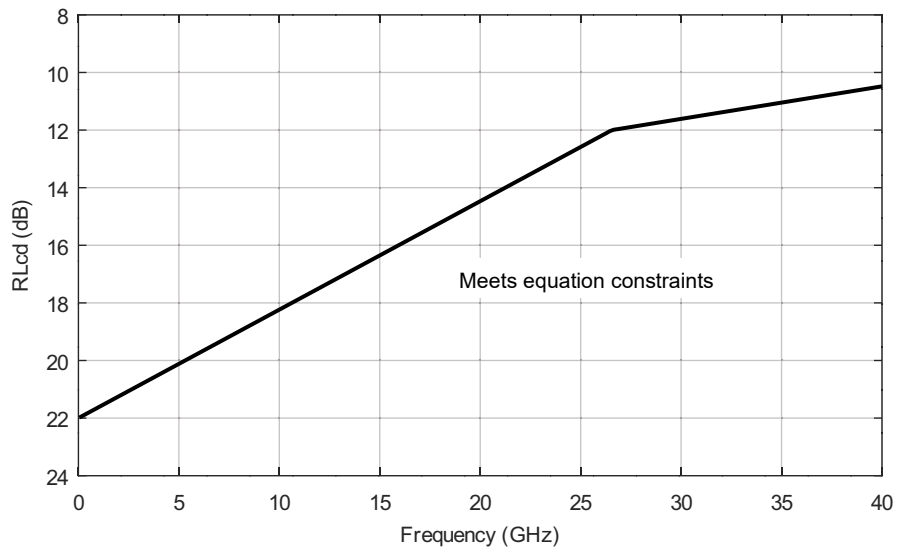


Figure 162–7—Receiver differential-mode to common-mode return loss

162.10 Channel characteristics

The channel is defined between TP0 and TP5 to include the transmitter and receiver differential controlled impedance PCB and the cable assembly as illustrated in Figure 162–2. The channel insertion loss, return loss, COM, and the transmitter and receiver differential controlled impedance PCB parameters are provided informatively in 162A.4 through 162A.7.

Channel definitions apply for links between two PHYs of the same type: 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4.

162.11 Cable assembly characteristics

Cable assemblies defined in this subclause contain insulated conductors terminated in a connector at each end for use as link segments between MDIs. Cable assemblies are primarily intended as point-to-point links between 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 PHYs using controlled impedance cables.

Three cable assembly types are specified:

- a) 100GBASE-CR1: Cable assembly that supports single-lane links between two 100GBASE-CR1 PHYs with achievable cable length of at least 2 m.
- b) 200GBASE-CR2: Cable assembly that supports two-lane links between two 200GBASE-CR2 PHYs with achievable cable length of at least 2 m.
- c) 400GBASE-CR4: Cable assembly that supports four-lane links between two 400GBASE-CR4 PHYs with achievable cable length of at least 2 m.

NOTE—It may be possible to construct compliant cable assemblies longer than indicated. Length of a cable assembly does not imply compliance to specifications.

100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 cable assembly types are described in Annex 162D. The MDIs are defined in Annex 162C.

For 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, the lanes are AC-coupled. The AC-coupling shall be within the cable assembly. It is recommended that it is within the plug connectors. It should be noted that there may be various methods for AC-coupling in actual implementations. The low-frequency 3 dB cutoff of the AC-coupling shall be less than 50 kHz. It is recommended that the value of the coupling capacitors be 100 nF. The capacitor limits the inrush charge and baseline wander.

All cable assembly measurements are to be made between TP1 and TP4 with cable assembly test fixtures as specified in Annex 162B. These cable assembly specifications are based upon twinaxial cable characteristics, but other cable types are acceptable if the specifications of this subclause are met.

Table 162–18 provides a summary of the cable assembly characteristics for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, and references to the subclauses addressing each parameter.

The specifications for the three cable assembly types are identical except the number of lanes.

162.11.1 Characteristic impedance and reference impedance

The nominal differential characteristic impedance of the cable assembly is 100 Ω . The differential reference impedance for cable assembly specifications shall be 100 Ω .

162.11.2 Cable assembly insertion loss

The measured insertion loss at 26.56 GHz of a cable assembly shall be less than or equal to 19.75 dB.

Table 162–18—Cable assembly characteristics summary

Description	Reference	Value	Unit
Insertion loss at 26.56 GHz, IL_{dd} (max)	162.11.2	19.75	dB
Insertion loss at 26.56 GHz, IL_{dd} (min)	162.11.2	11	dB
Minimum cable assembly ERL ^a	162.11.3	8.25	dB
Differential-mode to common-mode return loss, RL_{cd}	162.11.4	Equation (162–20)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	162.11.5	Equation (162–21)	dB
Common-mode to common-mode return loss, RL_{cc}	162.11.6	Equation (162–22)	dB
Minimum COM	162.11.7	3	dB

^a Cable assemblies with a COM greater than 4 dB are not required to meet minimum ERL.

The measured insertion loss of a cable assembly shall be greater than or equal to the minimum cable assembly insertion loss given in Equation (162–19) and illustrated in Figure 162–8.

$$IL_{dd}(f) \geq IL_{dd_{min}}(f) = \left\{ \begin{array}{ll} 0.418\sqrt{f} + 0.177f + 0.0059f^2 & 0.05 \leq f < 26.56 \\ 1.222\sqrt{f} + 0.138f + 0.0015f^2 & 26.56 \leq f \leq 40 \end{array} \right\} \quad (162-19)$$

where

$IL_{dd}(f)$ is the measured cable assembly insertion loss in dB

$IL_{dd_{min}}(f)$ is the minimum cable assembly insertion loss in dB

f is the frequency in GHz

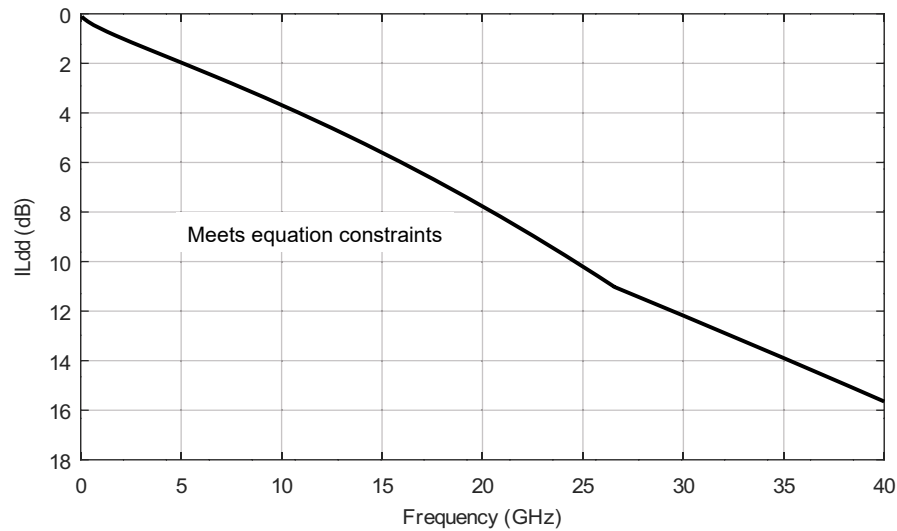


Figure 162–8—Minimum cable assembly insertion loss

162.11.3 Cable assembly ERL

The cable assembly ERL at TP1 and at TP4 is defined by the procedure in 93A.5 with the values in Table 162–19 and Table 162–20, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture cable-facing connection minus 0.2 ns.

The values of cable assembly ERL at TP1 and at TP4 shall meet the requirement specified in Table 162–18 for cable assemblies that have a COM less than 4 dB.

Table 162–19—Cable assembly ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	4500	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

162.11.4 Differential-mode to common-mode return loss

The cable assembly differential-mode to common-mode return loss shall meet Equation (162–20).

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (162-20)$$

where

$RLcd(f)$ is the cable assembly differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The cable assembly differential-mode to common-mode return loss limit is illustrated in Figure 162–9.

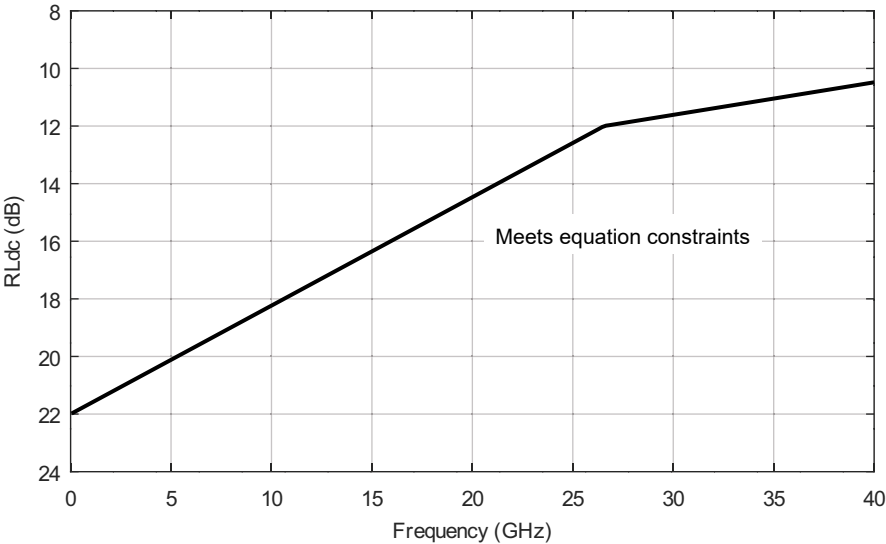


Figure 162-9—Cable assembly differential-mode to common-mode return loss

162.11.5 Differential-mode to common-mode insertion loss

The cable assembly differential-mode to common-mode insertion loss is specified relative to the insertion loss. The difference between the cable assembly differential-mode to common-mode insertion loss and the cable assembly insertion loss shall meet Equation (162-21) illustrated in Figure 162-10.

$$ILcd(f) - ILdd(f) \geq \begin{cases} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{cases} \quad (162-21)$$

where

- $ILcd(f)$ is the cable assembly differential-mode to common-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the cable assembly insertion loss at frequency f in dB
- f is the frequency in GHz

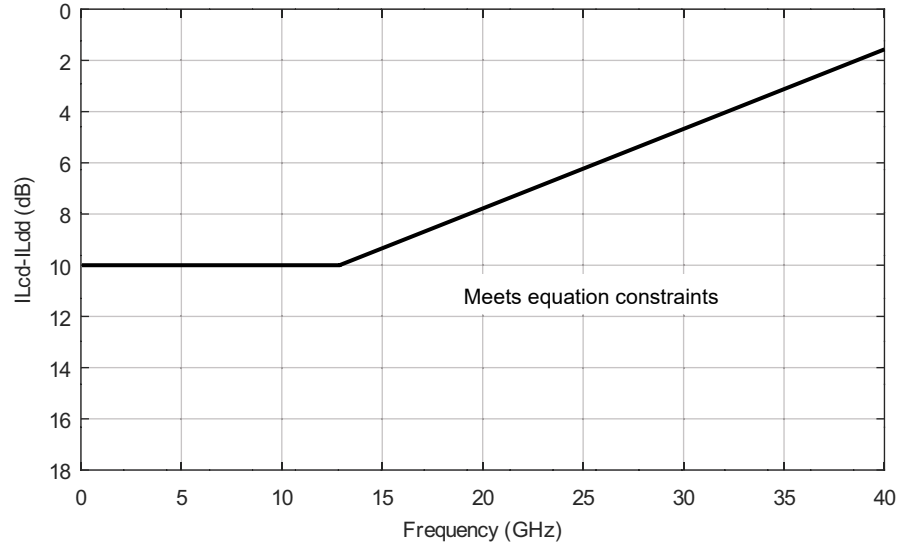


Figure 162-10—Cable assembly differential-mode to common-mode insertion loss to insertion loss difference

162.11.6 Common-mode to common-mode return loss

The cable assembly common-mode to common-mode return loss shall meet Equation (162-22).

$$RL_{cc}(f) \geq \left\{ \begin{array}{ll} 1.4 & 0.05 \leq f \leq 6 \\ 0.68 + 0.12f & 6 < f \leq 30 \\ 10.28 - 0.2f & 30 < f \leq 40 \end{array} \right\} \quad (162-22)$$

where

$RL_{cc}(f)$ is the cable assembly common-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The cable assembly common-mode to common-mode return loss limit is illustrated in Figure 162-11.

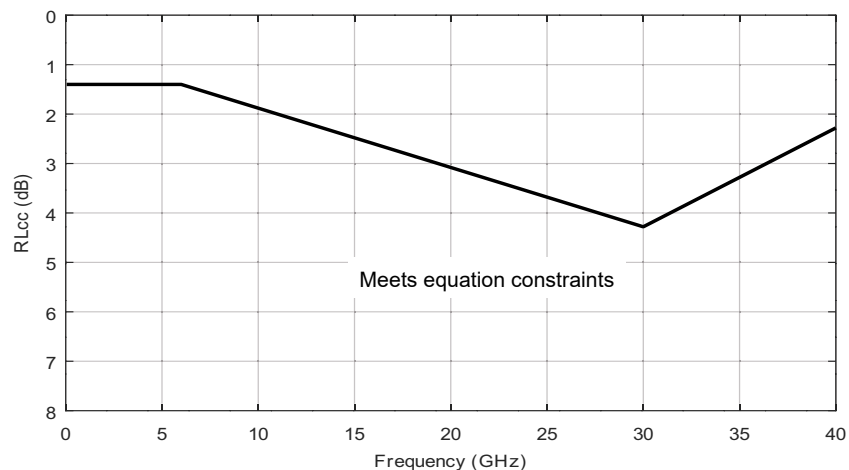


Figure 162–11—Cable assembly common-mode to common-mode return loss

162.11.7 Cable assembly Channel Operating Margin

The cable assembly Channel Operating Margin (COM) for each lane is derived from measurements of the cable assembly signal, near-end crosstalk, and far-end crosstalk paths. COM is computed using the path calculations defined in 162.11.7.1 and the procedure in 93A.1, where T_r is 7.5 ps for $H_t(f)$ as used in Equation (93A–19). The specific paths used depend on cable assembly types (see Annex 162D), as described in 162.11.7.2.

COM parameter values for the three cable assembly types are provided in Table 162–20.

COM shall be computed twice, Test 1 and Test 2, which differ in the value of the device package model transmission line length z_p . COM for any lane within the cable assembly shall be greater than or equal to 3 dB for both Test 1 and Test 2.

Table 162–20—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	1.2×10^{-4}	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1	z_p	12	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Transmission line characteristic impedance	Z_c	87.5	Ω
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line 2 characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient	$c(-3)$	—	—
Minimum value		–0.06	
Maximum value		0	
Step size		0.02	
Transmitter equalizer, 2 nd pre-cursor coefficient	$c(-2)$	—	—
Minimum value		0	
Maximum value		0.12	
Step size		0.02	
Transmitter equalizer, 1 st pre-cursor coefficient	$c(-1)$	—	—
Minimum value		–0.34	
Maximum value		0	
Step size		0.02	
Transmitter equalizer, post-cursor coefficient	$c(1)$	—	—
Minimum value		–0.2	
Maximum value		0	
Step size		0.02	
Continuous time filter, DC gain	g_{DC}	—	—
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Continuous time filter, DC gain 2	g_{DC2}	—	—
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz

Table 162–20—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage			
Victim	A_v	0.413	V
Far-end aggressor	A_{fe}	0.413	V
Near-end aggressor	A_{ne}	0.608	V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	32.5	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	12	UI
Normalized DFE coefficient maximum limit	$bb_{\max}(n)$		
for $n = 1$		0.85	—
for $n = 2$		0.3	—
for $n = 3$ to 6		0.2	—
for $n = 7$ to N_b		0.1	—
Normalized DFE coefficient minimum limit	$bb_{\min}(n)$		
for $n = 1$		0.3	—
for $n = 2$		0.05	—
for $n = 3$ to N_b		–0.03	—
Number of DFE floating tap banks	N_{bg}	3	—
Number of DFE floating taps per bank	N_{bf}	3	—
DFE maximum span including floating taps	N_f	40	UI
Normalized coefficient magnitude limit for DFE floating taps	b_{gmax}	0.03	—
DFE floating tap tail root-sum-of-squares limit	σ_{tmax}	0.02	—
DFE floating tap tail starting position	N_{ts}	25	UI
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	9×10^{-9}	V ² /GHz
Target detector error ratio	DER_0	10^{-4}	—

162.11.7.1 Channel signal and crosstalk path calculations

The channel paths between TP0 and TP5 used for calculation of the cable assembly COM consist of measured cable assembly signal and crosstalk paths, representative transmitter PCB signal paths, and representative receiver PCB signal paths.

The scattering parameters for a PCB transmission line are calculated using the method defined in 93A.1.2.3 with the parameter values given in Table 162–21. The PCB trace length parameter z_p has a different value for each specific signal path, as specified in 162.11.7.1.1 and 162.11.7.1.2.

The channel path calculations use the function `cascade()` defined in 93A.1.2.1.

Table 162–21—PCB model parameters and values

Parameter	Value	Units
γ_0	0	1/mm
a_1	3.8206×10^{-4}	ns ^{1/2} /mm
a_2	9.5909×10^{-5}	ns/mm
τ	5.79×10^{-3}	ns/mm
C_0	2.9×10^{-5}	nF
C_1	1.9×10^{-5}	nF
Z_c	100	Ω
R_0	50	Ω

162.11.7.1.1 Channel signal path

The scattering parameters of the channel signal path from TP0 to TP5 are calculated using Equation (162–25). The transmitter and receiver PCB signal paths are denoted as $S^{(HOSPT)}$ and $S^{(HOSPR)}$, and are calculated using Equation (162–23) and Equation (162–24), respectively. The PCB transmission line scattering parameters are denoted as $S^{(l)}$ and are calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 110.3$ mm in length and the parameter values given in Table 162–21, representing an insertion loss of 4.33 dB at 26.56 GHz on each PCB. The scattering parameters for the PCB via capacitances, $S^{(c0)}$ and $S^{(c1)}$, are calculated using Equation (93A–8) and the values in Table 162–21.

$$S^{(HOSPT)} = \text{cascade}(S^{(c0)}, S^{(l)}, S^{(c1)}) \quad (162-23)$$

$$S^{(HOSPR)} = \text{cascade}(S^{(c1)}, S^{(l)}, S^{(c0)}) \quad (162-24)$$

$$SCHS_p^{(k)} = \text{cascade}(S^{(HOSPT)}, S^{(CASP)}, S^{(HOSPR)}) \quad (162-25)$$

where

$SCHS_p^{(k)}$	is the channel signal path
$S^{(HOSPT)}$	is the host transmitter PCB signal path
$S^{(HOSPR)}$	is the host receiver PCB signal path
$S^{(CASP)}$	is the cable assembly signal path (TP1 to TP4)
k	is equal to zero

162.11.7.1.2 Channel crosstalk paths

The MDI is a significant contributor to crosstalk and is included in and characterized by the cable assembly crosstalk measurements. Crosstalk includes a near-end path where the aggressor is the PMD transmitter, and in some cases, additional near-end, far-end, and alien far-end crosstalk paths where the aggressors are other PMD transmitters that are connected to the same cable assembly.

For the channel crosstalk paths, the receiver PCB model is $S^{(HOSPR)}$ as defined in 162.11.7.1.1. The aggressor transmitter host PCB model is denoted as $S^{(HOTxSP)}$ and is calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 110.3$ mm in length and the parameter values given in Table 162–21, representing an insertion loss of 4.33 dB at 26.56 GHz.

The scattering parameters of the channel near-end crosstalk paths are calculated using Equation (162–26). The scattering parameters of the channel alien far-end crosstalk paths are calculated using Equation (162–27).

$$SCHNXT_p^{(k)} = \text{cascade}(S^{(HOTxSP)}, S^{(CANXTk)}, S^{(HOSPR)}) \quad (162-26)$$

where

$SCHNXT_p^{(k)}$	is the near-end crosstalk path
$S^{(HOSPR)}$	is the host receiver PCB signal path defined in 162.11.7.1.1
$S^{(HOTxSP)}$	is the aggressor transmitter PCB signal path
$S^{(CANXTk)}$	is the cable assembly near-end crosstalk path k (TP1 to TP4)
k	is the index of the near-end crosstalk path

$$SCHFXT_p^{(k)} = \text{cascade}(S^{(HOTxSP)}, S^{(CAFXTk)}, S^{(HOSPR)}) \quad (162-27)$$

where

$SCHFXT_p^{(k)}$	is the alien far-end crosstalk path
$S^{(HOSPR)}$	is the host receiver PCB signal path defined in 162.11.7.1.1
$S^{(HOTxSP)}$	is the aggressor transmitter PCB signal path
$S^{(CAFXTk)}$	is the cable assembly far-end crosstalk path k (TP1 to TP4)
k	is the index of the alien far-end crosstalk path

162.11.7.2 Signal and crosstalk paths used in calculation of COM

Cable assemblies have several types, as described in Annex 162D. The choice of signal and crosstalk paths for calculation of COM is specific to each cable assembly type, as specified in Table 162–22.

The signal path is calculated using Equation (162–25).

The near-end crosstalk paths are calculated using Equation (162–26), with k values from 1 to n ; where n is the number of near-end crosstalk paths given in Table 162–22 for the cable assembly type considered.

The far-end crosstalk paths are calculated using Equation (162–27), with k values from 1 to n ; where n is the number of far-end crosstalk paths given in Table 162–22 for the cable assembly type considered.

The specifications for the cable assembly types are identical except the number of lanes.

The number of crosstalk paths based on MDI type are given in Table 162–22.

Table 162–22—Number of crosstalk paths used in COM

Victim (one end)	NEXT	FEXT (other end)			
		SFP112	SFP-DD112 or DSFP	QSFP112	QSFP-DD800 or OSFP
SFP112	1	0	1	3	7
SFP-DD112 or DSFP	2	1	1	3	7
QSFP112	4	3	3	3	7
QSFP-DD800 or OSFP	8	7	7	7	7

162.12 MDI specifications

The MDI couples the PMD (specified in 162.8 and 162.9) to the cable assembly (specified in 162.11).

100GBASE-CR1 has six specified MDI connectors: SFP112, SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

200GBASE-CR2 has five specified MDI connectors: SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

400GBASE-CR4 has three specified MDI connectors: QSFP112, QSFP-DD800, and OSFP.

Annex 162C specifies the MDIs for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4.

162.13 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 14.7.

162.14 Protocol implementation conformance statement (PICS) proforma for Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4¹⁸

162.14.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

162.14.2 Identification

162.14.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

162.14.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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¹⁸Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

162.14.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CR1	100GBASE-CR1 PMD	162.1	Can operate as a 100GBASE-CR1 PMD	O.1	Yes [] No []
*CR2	200GBASE-CR2 PMD	162.1	Can operate as a 200GBASE-CR2 PMD	O.1	Yes [] No []
*CR4	400GBASE-CR4 PMD	162.1	Can operate as a 400GBASE-CR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-P PMA	162.1	Device implements Clause 135 PMA for 100GBASE-P	CR1:M	Yes [] N/A []
PMA200	200GBASE-R PMA	162.1	Device implements Clause 120 PMA for 200GBASE-R	CR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	162.1	Device implements Clause 120 PMA for 400GBASE-R	CR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	162.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	162.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!CR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	162.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	162.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC	162.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	CR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	162.1	Device implements Clause 82 100GBASE-R PCS	CR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	162.1	Device implements Clause 119 200GBASE-R PCS	CR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	162.1	Device implements Clause 119 400GBASE-R PCS	CR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	162.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	162.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	162.6	Conforms to constraints	M	Yes []
*MD	MDIO capability	162.7	Device implements Clause 45 MDIO with the variable mapping in 162.7.	O	Yes [] No []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CBL	Cable assembly	162.11	Item marked with CBL are cable assembly specifications, these items are the only ones applicable to cable assemblies and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	162.9.4		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

162.14.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162.14.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

162.14.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed_ <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	162.9.4.1.3	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []
PC9	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC10	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []

162.14.4.3 Transmitter specifications

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	162.9.4	Per Table 162–11	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD is not in same package as PCS	162.9.4	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	Output voltage specifications	93.8.1.3	Per Table 162–11	M	Yes []
TC4	ERL	162.9.4.8	Per Table 162–11	M	Yes []
TC5	Common-mode to differential-mode return loss	162.9.4.10	Per Table 162–11	M	Yes []
TC6	Common-mode to common-mode return loss	162.9.4.9	Per Table 162–11	M	Yes []
TC7	Steady-state voltage, v_f	162.9.4.1.2	Per Table 162–11	M	Yes []
TC8	Linear fit pulse peak (min)	162.9.4.1.2	Per Table 162–11	M	Yes []
TC9	Level separation mismatch ratio	162.9.4	Per Table 162–11	M	Yes []
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 162–11	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Absolute change in coefficients other than $c(\text{coef_sel})$	162.9.4.1.4	Per Table 162–11	M	Yes []
TC12	Coefficient range	162.9.4.1.5	Per Table 162–11	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120D.3.1.6	Per Table 162–11	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	162.9.4	Per Table 162–11	M	Yes []
TC15	Output jitter	162.9.4.7	Per Table 162–11	M	Yes []
TC16	Low-frequency AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 162–11	M	Yes []
TC17	Full-band AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 162–11	M	Yes []

162.14.4.4 Receiver specifications

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Signaling rate range	162.9.5.1	Per Table 162–15	M	Yes []
RC2	Differential-mode to common-mode return loss at TP3	162.9.5.6	Per Table 162–15	M	Yes []
RC3	Input amplitude tolerance	162.9.5.2	Meets FEC symbol error ratio requirement under described conditions	M	Yes []
RC4	Interference tolerance	162.9.5.3	PHY meets FEC symbol error ratio requirement in all tests defined in Table 162–16 with the specified test setup and procedure	M	Yes []
RC5	Jitter tolerance	162.9.5.4	PHY meets FEC symbol error ratio requirement with the specified test setup and procedure	M	Yes []
RC6	ERL	162.9.5.5	Per Table 162–15	M	Yes []

162.14.4.5 Cable assembly specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CA1	Differential reference impedance	162.11.1	100 Ω	CBL:M	Yes [] N/A []
CA2	Minimum insertion loss	162.11.2	Per Equation (162–19)	CBL:M	Yes [] N/A []
CA3	Maximum insertion loss at 26.56 GHz	162.11.2	19.75 dB	CBL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CA4	ERL	162.11.3	Greater than or equal to 8.25 dB for cable assemblies that have a COM less than 4 dB	CBL:M	Yes [] N/A []
CA5	Differential-mode to common-mode input and output return loss	162.11.4	Per Equation (162–20)	CBL:M	Yes [] N/A []
CA6	Differential-mode to common-mode insertion loss	162.11.5	Per Equation (162–21)	CBL:M	Yes [] N/A []
CA7	Common-mode to common-mode return loss	162.11.6	Per Equation (162–22)	CBL:M	Yes [] N/A []
CA8	Cable assembly Channel Operating Margin (COM)	162.11.7	Greater than or equal to 3 dB for both Test 1 and Test 2, for any lane within the cable assembly	CBL:M	Yes [] N/A []
CA9	AC-coupling	162.11	3 dB cutoff frequency less than 50 kHz	CBL:M	Yes [] N/A []

162.14.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Environmental specifications	162.13	Conform to applicable requirements of 14.7	M	Yes []